TRUSTCHARGE: AI-BASED CENTRALIZED EV CHARGING RELIABILITY AND SMART RESERVATION PLATFORM

Project Phase I Report

submitted to the APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the award of the degree

Bachelor of Technology

in

Computer Science and Engineering

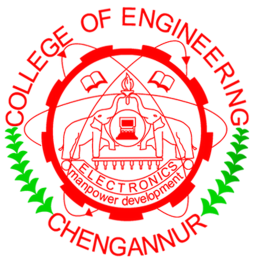
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CERTIFICATE

This is to certify that the project report titled ***TRUSTCHARGE: AI-BASED CENTRALIZED EV CHARGING RELIABILITY AND SMART RESERVATION PLATFORM*** *is a bonafide record of the* ***CSD 415 PROJECT PHASE I*** *presented by* ***ALEESHA M A (CHN23CS030), JIA CHINNU ABRAHAM (CHN23CS101), JOEL GEORGE MATHEW (CHN23CS102) & LOIS ROSE DAVID (CHN23CS120)****, Seventh Semester B.Tech. students in Computer Science and Engineering, under our guidance and supervision. This report is submitted in partial fulfillment of the requirements for the award of the degree* ***B.Tech. in Computer Science and Engineering*** *of* ***APJ Abdul Kalam Technological University****.*

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DECLARATION

We undersigned hereby declare that the project report **"TrustCharge: AI-Based Centralized EV Charging Reliability and Smart Reservation Platform"**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under the supervision of **Prof. Gopakumar G**, Department of Computer Engineering. This submission represents our ideas in our own words, and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma, or similar title of any other University.

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ABSTRACT

Electric vehicle adoption increases the need for dependable and accessible charging infrastructure. However, charging information is fragmented across provider-specific applications, and users may have difficulty comparing stations, checking availability, planning a route, and securing a charging bay. TrustCharge is a centralized EV charging platform intended to bring station discovery, vehicle information, navigation, reservation, wallet payments, and membership benefits into one web application. The implemented system aggregates station information from NDW, TomTom, and OpenChargeMap, uses RDW open data for vehicle verification, and supports Smartcar OAuth integration for vehicle telemetry. The frontend is built with React and Tailwind CSS, while the backend uses Node.js and Express with SQLite persistence. Firebase Authentication supports identity management; Mollie supports checkout and payment flows; Google Gemini provides an AI assistant with a deterministic local fallback; and OpenStreetMap services support map and routing features. The system includes station discovery, station ratings and fault reports, wallet top-ups, membership plans, bay reservation, cancellation and refund handling, and trip simulation with wallet-based energy billing. The project presentation reports 1,574 stations tracked and 3,990 observation samples logged across nine external services. The collected observations currently report UNKNOWN station status, so a trained AI reliability model and queue-time prediction model are retained as future work rather than presented as completed functionality. TrustCharge establishes an extensible foundation for a unified EV charging experience.

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ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
BEV	Battery Electric Vehicle
CSS	Cascading Style Sheets
ETA	Estimated Time of Arrival
EV	Electric Vehicle
G2V	Grid-to-Vehicle
HTTP	Hypertext Transfer Protocol
ML	Machine Learning
NDW	National Data Warehouse for Traffic Information (Netherlands)
OCM	OpenChargeMap
OSM	OpenStreetMap
OAuth	Open Authorization
RDW	Netherlands Vehicle Authority
REST	Representational State Transfer
SRS	Software Requirements Specification
UI	User Interface
V2G	Vehicle-to-Grid

CHAPTER 1

INTRODUCTION

The increasing adoption of Electric Vehicles (EVs) has created a growing demand for reliable and accessible charging infrastructure. Unlike conventional fuel stations, electric vehicle charging stations may have different connector types, charging speeds, tariffs, and real-time availability conditions. Finding a suitable charging station and ensuring that a charging port is available at the required time can therefore be challenging for EV users. In addition, the lack of centralized information about charging station availability and reliability can lead to unnecessary travel, waiting time, and range anxiety.

Existing EV charging platforms provide features such as charging station discovery, navigation, and charging information. However, users may still face difficulties in identifying reliable stations, checking live charging availability, planning trips based on vehicle range, and reserving charging ports in advance. Differences in charging network data sources and incomplete availability information can further affect the reliability of station selection. These challenges highlight the need for a centralized platform that combines charging station information, reservation facilities, navigation, and intelligent decision support.

Artificial Intelligence (AI), real-time data integration, and web technologies provide opportunities to improve the EV charging experience. By combining information from charging station networks, location services, and vehicle details, a centralized platform can help users identify suitable charging stations and plan their journeys more effectively. Membership-based services and digital wallet integration can also simplify reservations and provide additional benefits to frequent EV users.

This project, **TrustCharge: AI-Based Centralized EV Charging Reliability and Smart Reservation Platform**, develops a web-based platform designed to improve the reliability and accessibility of EV charging services. The system integrates charging station discovery, availability information, smart trip planning, charging port reservation, membership management, and wallet-based payments. It uses external charging station and mapping services to support station discovery and navigation. The platform also provides vehicle-related information and AI-assisted decision support to help users make informed charging decisions. TrustCharge

aims to offer a centralized and user-friendly solution for EV charging management.

1.1 Project Area

The project lies in the fields of Artificial Intelligence, web application development, electric vehicle infrastructure management, location-based services, and database management. It combines a React.js frontend with a Node.js and Express backend to provide an interactive interface for discovering charging stations, managing reservations, and accessing user account information. MySQL is used for structured data management, while Firebase Authentication supports user authentication.

TrustCharge integrates external services such as OpenChargeMap and OpenStreetMap for charging station information and map-based navigation. Additional charging network data sources are used to obtain station information and availability where supported. The platform includes an intelligent trip-planning interface, charging port reservation functionality, membership-based benefits, and wallet management. These features are intended to improve the overall charging experience by bringing essential EV charging services together in one centralized application.

1.2 Objectives

The main objectives of TrustCharge are:

1. **Centralized charging station discovery:** To provide a unified platform for discovering EV charging stations and viewing their available information.

2. **Charging availability monitoring:** To integrate external charging network data sources to display station and port availability wherever reliable data is available.

3. **Smart trip planning:** To help users plan journeys by considering destination, vehicle information, estimated driving distance, and charging requirements.

4. **Membership-based priority access:** To provide membership-based benefits and priority reservation options according to the configured membership rules.

5. **Secure user authentication:** To provide user registration and authentication facilities to protect access to personal accounts and reservation-related features.

6. **Centralized data management:** To maintain structured information related to users, vehicles, charging stations, reservations, membership plans, and wallet transactions.

7. **Reservation and transaction tracking:** To maintain reservation and transaction records so that users can review their charging-related activities.

8. **Responsive interface:** To design a responsive web interface that can be accessed across different screen sizes and devices.

9. **External API integration:** To integrate charging station, mapping, navigation, and payment services through appropriate external APIs.

10. **Charging port reservation:** To allow users to select charging stations, choose suitable charging ports, and reserve charging slots in advance.

11. **Reliability-based station selection:** To help users compare charging stations using available information such as distance, charging power, availability, and ratings.

12. **Membership management:** To provide membership plans with benefits such as reservation privileges, charging discounts, and wallet credits.

13. **Wallet and payment integration:** To support wallet management and integrate payment services for applicable membership and charging-related transactions.

14. **Vehicle integration:** To maintain vehicle-related information and incorporate supported vehicle data into trip planning and charging assistance.

15. **AI-assisted decision support:** To provide intelligent assistance for charging-related queries and help users make informed decisions using available platform information.

16. **User-friendly web application:** To develop a responsive and accessible interface that brings charging station discovery, navigation, reservations, membership, and payment functionality together.

CHAPTER 2

PROBLEM DEFINITION AND MOTIVATION

2.1 Existing System

Existing EV charging services commonly provide station maps, navigation, availability information, and payment options. Examples discussed in the project materials include Tata Power EZ Charge, Statiq, ChargeZone, and Bolt.Earth. These services are useful within their supported networks, but information and account features can be distributed across independent applications. A driver may have to compare several sources to find a station that matches the vehicle, route, time, and charging needs.

2.1.1 Limitations of the Existing System

The project identifies the following limitations:

- Users may need separate applications for different charging providers.
- Station information and availability may be inconsistent or incomplete across sources.
- There is no single cross-network view for comparing station distance, charging power, tariff, and reported availability.
- Membership benefits, reservation rules, and wallet balances are not unified across independent providers.
- Queue-time prediction and charger reliability prediction are not consistently available to users.
- Users may travel to a station that is listed as available but cannot provide a satisfactory charging experience.

2.2 Problem Statement

To address the unreliability and fragmentation of EV charging information, there is a need for a centralized platform that aggregates information from multiple providers and supports station comparison, route guidance, reservation, and unified wallet and membership services. The

platform should make the status and limitations of the available data clear, while providing a basis for future predictive reliability and waiting-time services.

2.3 Need for the Proposed System

A centralized interface can reduce the effort required to search across separate services. By combining station discovery with vehicle information, route support, reservation, and payment-related features, TrustCharge aims to make charging decisions more convenient. A unified workflow can also help users understand estimated cost and membership benefits before reserving a charging bay.

2.4 Proposed Solution

TrustCharge uses a React-based frontend and a Node.js/Express backend to connect users with aggregated charging-station data and project services. The platform integrates NDW, TomTom, and OpenChargeMap for station-related information; RDW data for vehicle verification; Smartcar for optional vehicle telemetry; Firebase Authentication for identity; Mollie for payment checkout; and OpenStreetMap services for maps and routing. Reservation, wallet, membership, trip simulation, and assistant features are integrated into the same application. Predictive charger-health and queue-time models are described as future extensions because the current observations do not yet provide validated model outputs.

CHAPTER 3

LITERATURE REVIEW

3.1 Multi-Objective EV Charging Station Allocation (2023)

Multi-objective allocation research considers electrical and planning measures such as power loss, voltage profile, operational cost, and renewable-energy utilization. The reviewed work uses the IEEE 69-bus system to evaluate charging-station and renewable-energy placement [1]. Such methods help infrastructure planners decide where stations should be placed, but they do not directly provide user-facing real-time booking, predictive availability, or a unified wallet.

3.2 Integration of Electric Vehicles into Smart Grids (2024)

Smart-grid studies investigate coordination between grid-to-vehicle (G2V) charging and vehicle-to-grid (V2G) energy transfer. The reviewed approach applies an Improved Honey Badger Algorithm to EV energy management and evaluates efficiency, power loss, and total harmonic distortion [2]. These objectives are important for grid operation, but the work does not focus on navigation, cross-network reservation, or prepaid booking.

3.3 Smart Charging Communication Using ISO 15118 (2024)

Standards-based communication can support interoperability between vehicles and charging infrastructure. A reviewed study demonstrates ISO 15118-based smart charging at a public charging hub [3]. Its focus is charging control and communication rather than station discovery, route planning, reservation, or reliability prediction.

3.4 AI-Based EV Smart Charging System in Malaysia (2024)

The reviewed Malaysian study combines machine learning, predictive analytics, demand response, dynamic pricing, V2G integration, and mathematical optimization. The literature review reports 30% energy savings and 20.38% cost reduction [4]. The study also identifies limitations related to real-world deployment, localized datasets, and large-scale implementation. These findings support the potential value of data-driven charging, while emphasizing the need for appropriate validation.

3.5 AI-Driven Load Balancing and Grid Integration (2025)

A reviewed approach combines predictive demand forecasting, machine learning and deep learning, dynamic load balancing, demand response, smart-grid integration, and V2G. The reviewed material reports a 20% reduction in peak-period overload [5]. Its limitations include scalability and centralized-system vulnerabilities. Grid-level load balancing is complementary to, but different from, a consumer-facing system for station comparison and reservation.

3.6 Reliability-Based Planning for EVs and Renewable Energy (2025)

Reliability-based planning plays an important role in the development of electric vehicle charging infrastructure, particularly when charging demand increases and renewable energy sources are integrated into the electrical grid. The reviewed study combines Gated Recurrent Unit (GRU)-based electric vehicle load forecasting with an enhanced Harris Hawk Optimization method and a bus reliability index. The GRU model is used to estimate future EV charging loads, while the optimization method supports the identification of suitable planning solutions based on the considered system constraints. The bus reliability index is incorporated to evaluate the reliability of the electrical network and understand the impact of charging demand on system performance.The study evaluates different reliability and electrical-system measures and reports a reduction in the expected energy not supplied for the system under investigation [6]. This result highlights the importance of considering both charging demand and electrical-system reliability when planning EV charging infrastructure. Forecasting future demand can help infrastructure planners estimate the charging capacity required at different locations and reduce the risk of insufficient supply.

3.7 Reservation-Based Charging (2023)

Reservation-based charging is an approach designed to improve the coordination between EV users and charging station operators. Instead of travelling to a station without knowing whether a charging port will be available, users can request a charging slot in advance. The reviewed study investigates reservation methods that distribute charging requests among stations using real charging-station data [7]. Such methods can help organize charging demand and improve the utilization of available charging infrastructure.

The reservation process generally involves identifying a suitable charging station, selecting an

available time slot, and submitting a booking request. Depending on the system design, reservation mechanisms may consider station availability, charging requirements, and user preferences when assigning charging slots. By coordinating charging requests in advance, these methods can reduce uncertainty for drivers and help prevent excessive demand at individual stations. They can also improve the planning of charging activities, particularly in areas where the number of charging ports is limited.

3.8 Route Planning and Charging Reservation (2018)

Route planning is an important component of electric vehicle travel because EVs have limited driving ranges and may require charging stops during longer journeys. Selecting a suitable route involves more than identifying the shortest distance between the starting point and the destination. Drivers may also need to consider traffic conditions, charging station locations, charging costs, battery range, and the availability of charging facilities along the route.

3.9 Business Intelligence for EV Charging (2025)

Business-intelligence approaches analyze EV and charging-station data to recommend charging schedules. The reviewed work reports improvements in waiting time, station utilization, cost, and schedule recommendations [9]. Its challenges include the need for large quantities of real-time data, secure data sharing, and scalability.

3.10 Waiting-Time and Occupancy Prediction (2025)

Research on long-trip charging planning applies Mixed-Integer Dynamic Programming with Monte Carlo simulation to forecast waiting time, reporting reductions in total trip time and charging waiting time [10]. Adaptive federated learning has also been proposed for charging-station occupancy prediction using measures such as MAE, RMSE, and MAPE [11]. These approaches depend on the quality and representativeness of data, and further evaluation under diverse real-time conditions is important.

3.11 Charging Infrastructure Reliability Using Reviews (2023)

A cross-lingual deep-learning approach analyzes charging-station reviews using BERT-based methods and machine translation. The reviewed study reports 89.1% accuracy and an F1-score of 0.78 for identifying station problems [12]. User-generated reports can complement status data, but translated reviews and limited real-time support remain challenges.

CHAPTER 4

REQUIREMENT ANALYSIS

4.1 Hardware Requirements

The application can be developed and demonstrated on a standard laptop or desktop computer with a modern multi-core processor, at least 8 GB RAM recommended for development tools, sufficient storage for dependencies and project data, and an internet connection. Users need a computer or smartphone with a modern web browser. External API and payment integrations require network connectivity.

4.2 Software Requirements

The frontend uses React and Tailwind CSS. The backend uses Node.js and Express and stores persistent application data in SQLite. Firebase Authentication supports user identity. Mollie is used for checkout and payment flows. Google Gemini provides the AI assistant, with a deterministic local fallback. NDW, TomTom, OpenChargeMap, RDW, Smartcar, and OpenStreetMap-related services provide data or integration capabilities. Development requires a current Node.js runtime, npm, a code editor, and a supported web browser.

4.3 Functional Requirements

1. The system shall support user authentication and maintain a user profile.
2. The system shall allow vehicle information to be registered and verified using supported RDW data where applicable.
3. The system shall support optional Smartcar authorization for vehicle telemetry, subject to permissions and service availability.
4. The system shall aggregate charging-station information from supported external sources and display station details on a map and in searchable lists.
5. The system shall allow users to inspect station distance, charging power, available status information, tariff information when supplied, and user feedback where available.
6. The system shall support station ratings and fault reports.

7. The system shall provide wallet top-up and membership checkout workflows through the configured payment integration.

8. The system shall allow users to select a charging station, choose an arrival time and duration, select an available bay, and manage a reservation.

9. The system shall apply membership-based reservation rules, discounts, and cancellation conditions as configured by the application.

10. The system shall support map-based route guidance and trip simulation with estimated charging cost and wallet-based energy billing.

11. The system shall provide an AI assistant for charging-related questions and use a local fallback when the external assistant is unavailable.

4.4 Non-Functional Requirements

Usability: The interface should be responsive and present station, vehicle, wallet, and reservation information clearly.

Performance: Common user actions should complete promptly under normal network and service conditions.

Security: Authentication, payment redirects, and external credentials must be handled through the appropriate services; secrets should not be exposed in client-side code.

Reliability: The application should handle unavailable external services gracefully and distinguish unavailable data from confirmed status.

Maintainability: Provider integrations and application modules should be organized so that new data sources and regions can be added.

Scalability: The design should allow future improvements to data aggregation, reservation processing, and predictive services.

Privacy: Vehicle telemetry and location access should be used only after the user grants the relevant permissions.

CHAPTER 5

DESIGN AND IMPLEMENTATION

TrustCharge is a centralized electric vehicle charging platform designed to integrate charging station discovery, reliability information, vehicle verification, trip planning, charging reservations, wallet management, and membership services. The system follows a web-based architecture consisting of a React frontend, a Node.js and Express backend, a SQLite database, and external services for authentication, charging station information, vehicle telemetry, payments, mapping, and AI-assisted recommendations.

This chapter presents the system design, including the use case diagram, data flow diagrams, system architecture, and implementation methodology.

5.1 Design

5.1.1 Use Case Diagram

The use case diagram illustrates the interactions between the users and the TrustCharge platform. The primary actors are the EV Driver and the Fleet Owner. EV drivers can authenticate themselves, verify their vehicles, view charging stations, obtain AI-assisted recommendations, plan routes, manage their wallets, purchase memberships, and reserve charging bays. Fleet owners can additionally manage fleet vehicles, monitor vehicle telemetry, reserve charging bays for fleet vehicles, and review fleet charging sessions and costs.

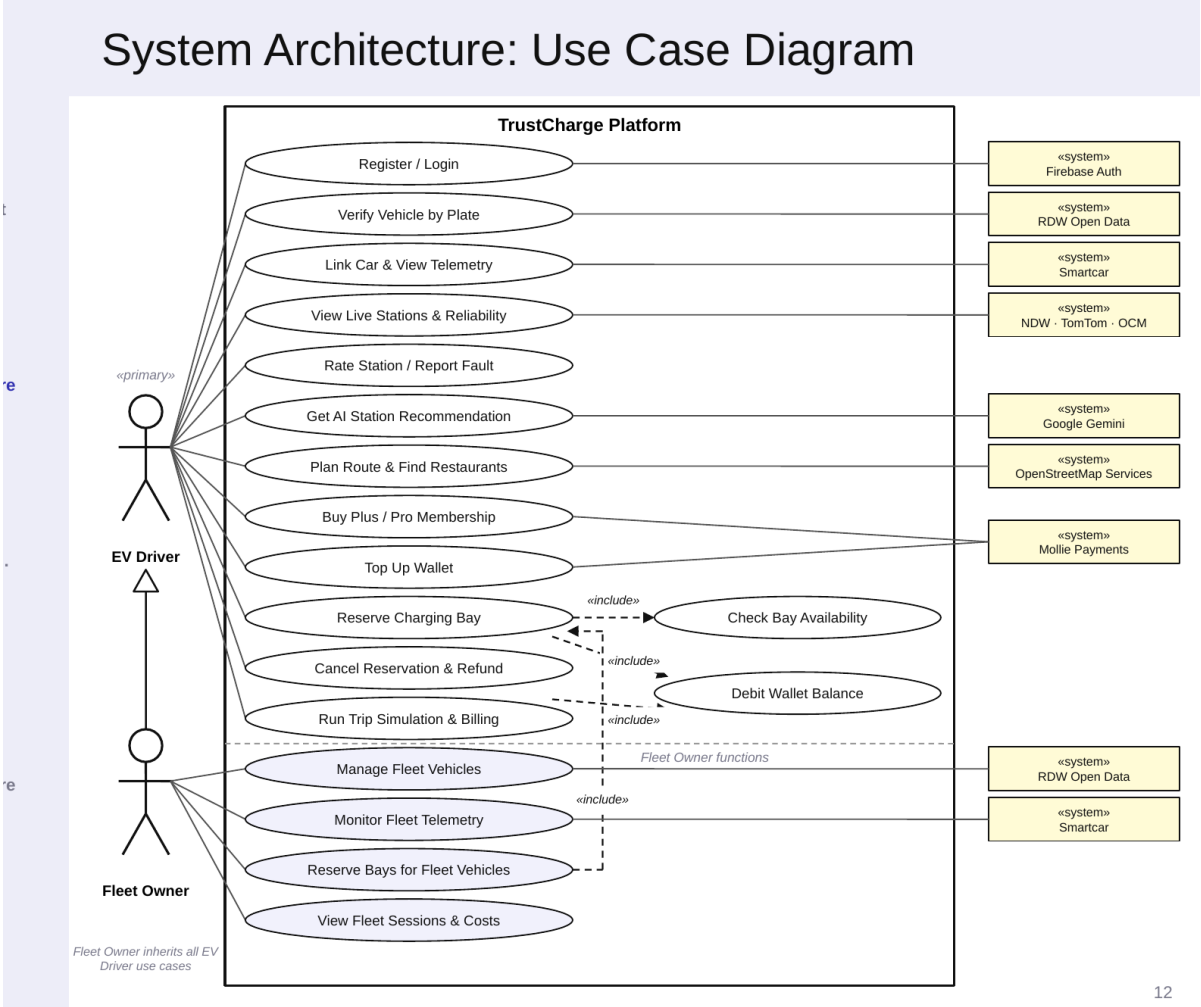


Figure 5.1: Use Case Diagram of TrustCharge

5.1.2 Data Flow Diagram

5.1.2.1 Level-0 Data Flow Diagram

The Level-0 data flow diagram represents TrustCharge as a single centralized process. EV drivers and fleet owners interact with the platform to request charging station information, obtain recommendations, manage reservations, and perform payment-related operations. The platform exchanges data with external services for authentication, vehicle verification, telemetry, charging station information, payments, AI recommendations, and navigation.

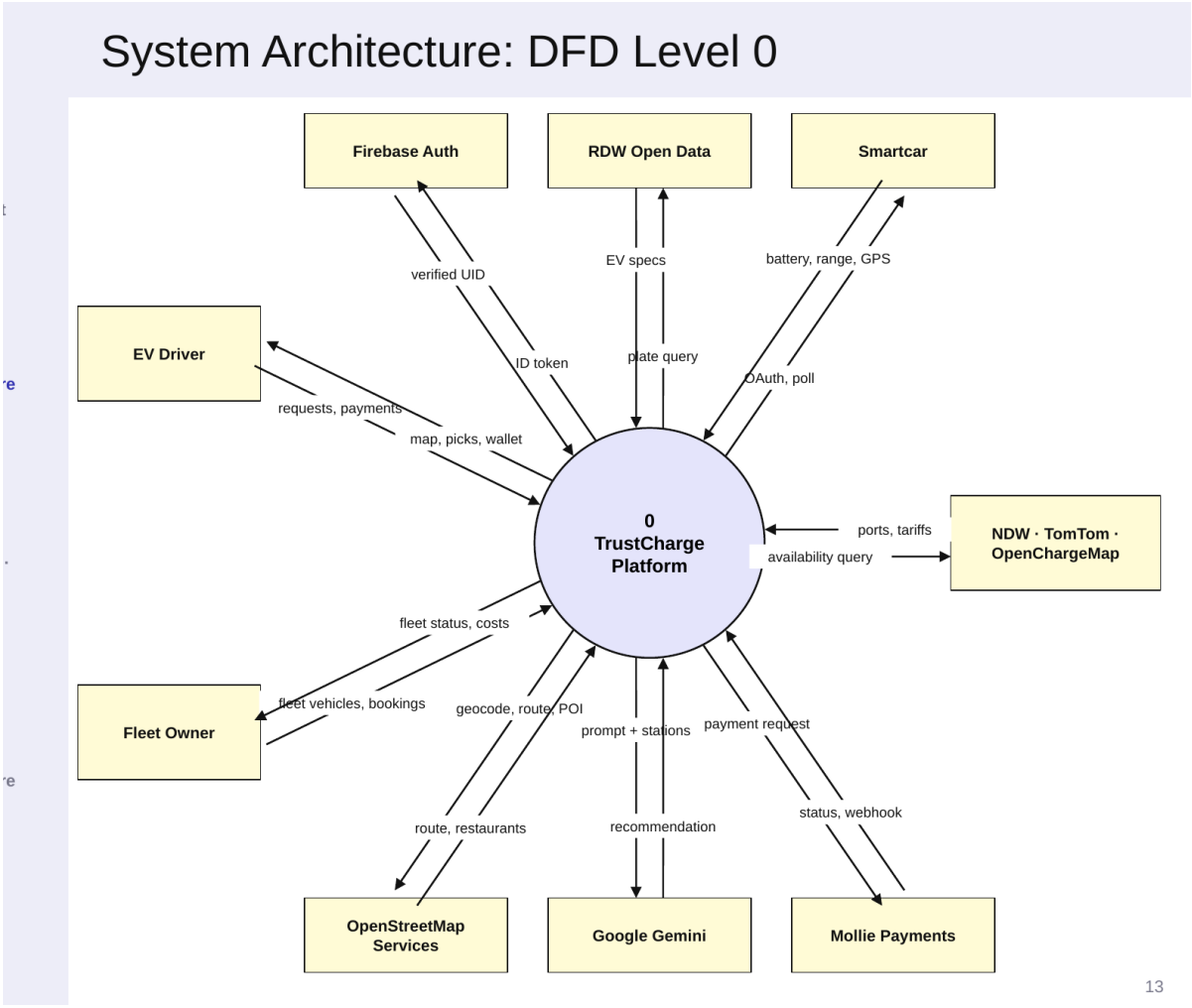


Figure 5.2: Level-0 Data Flow Diagram of TrustCharge

5.1.2.2 Level-1 Data Flow Diagram: Identity and Vehicle

The first Level-1 data flow diagram describes user authentication, vehicle verification, and vehicle telemetry. Firebase Authentication supports user authentication, while RDW Open Data is used for vehicle verification. Smartcar integration supports access to vehicle telemetry, including battery level, estimated range, and location, where available.

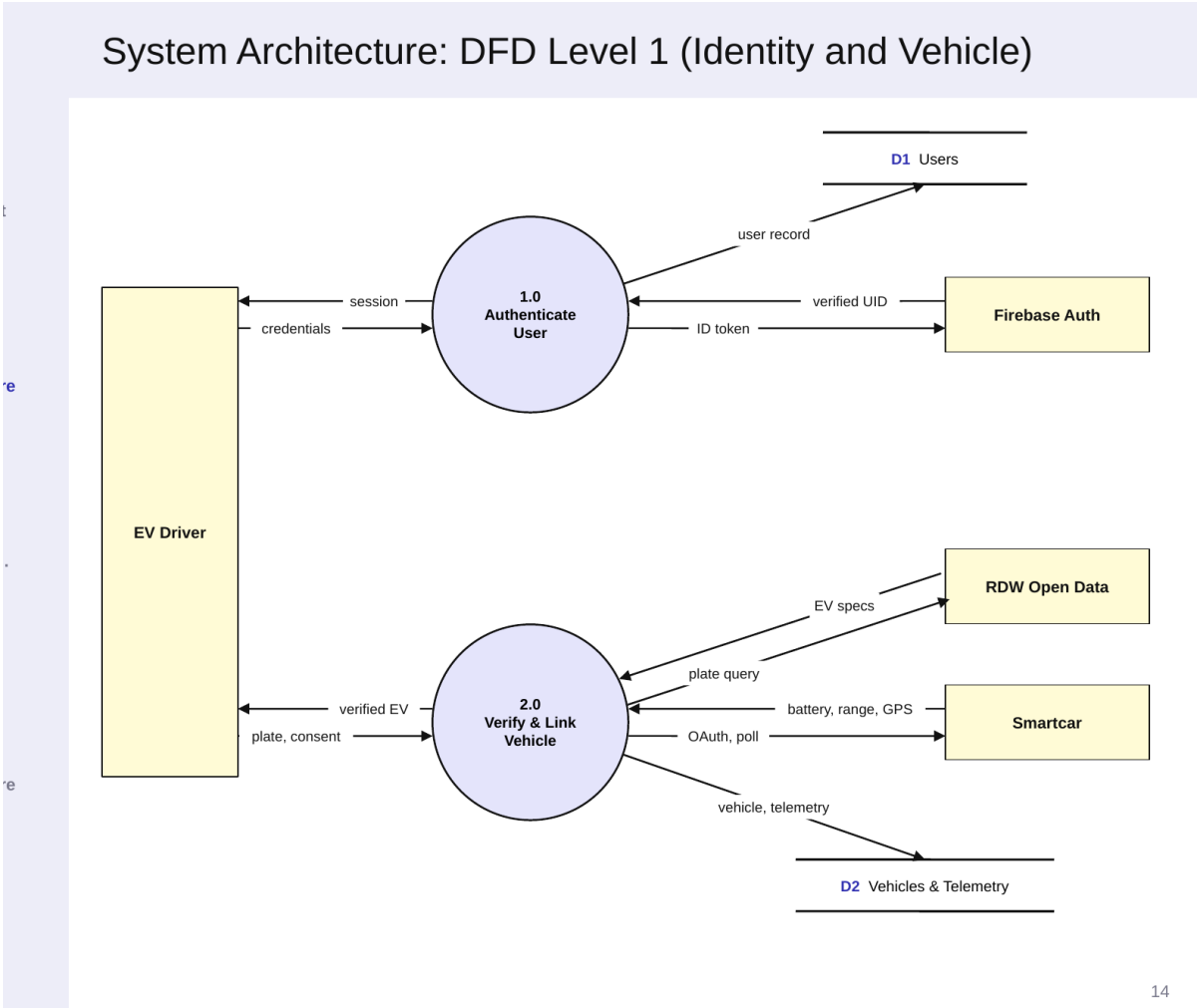


Figure 5.3: Level-1 DFD: Identity and Vehicle

5.1.2.3 Level-1 Data Flow Diagram: AI and Charging Stations

This diagram describes charging station aggregation, reliability scoring, station recommendations, and route planning. Station information is obtained from integrated sources such as NDW, TomTom, and OpenChargeMap. Station observations, ratings, and reports are used by the reliability component. Google Gemini supports AI-assisted station recommendations, while OpenStreetMap services support geocoding, routing, and nearby-point-of-interest discovery.

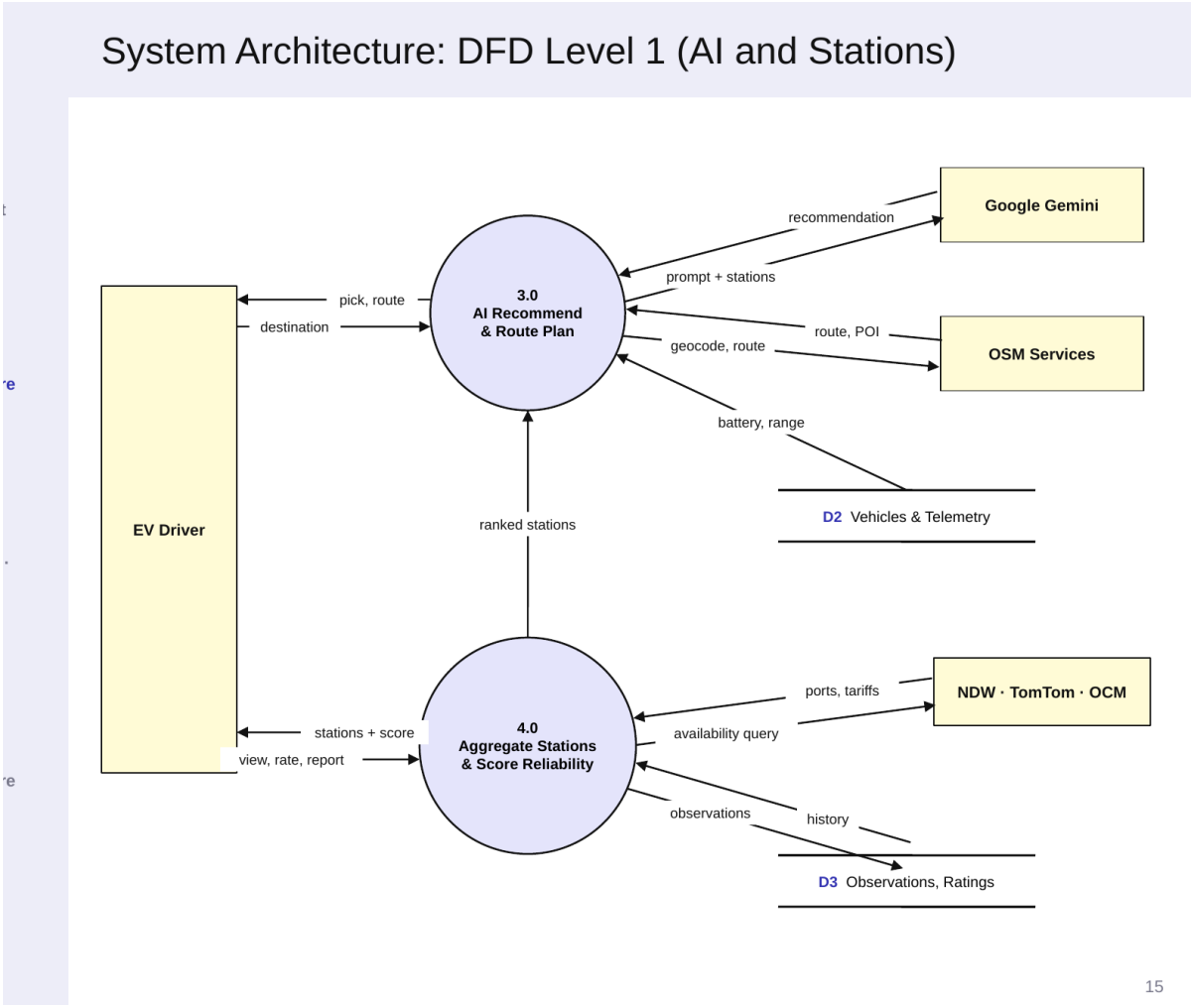


Figure 5.4: Level-1 DFD: AI and Charging Stations

5.1.2.4 Level-1 Data Flow Diagram: Wallet and Billing

This diagram represents wallet management, membership purchases, charging bay reservations, trip simulation, and billing. Users can manage their wallet and membership plans, reserve charging bays, and cancel eligible reservations. Mollie supports applicable payment transactions, while the backend manages reservation records, wallet balances, refunds, and charging-related costs.

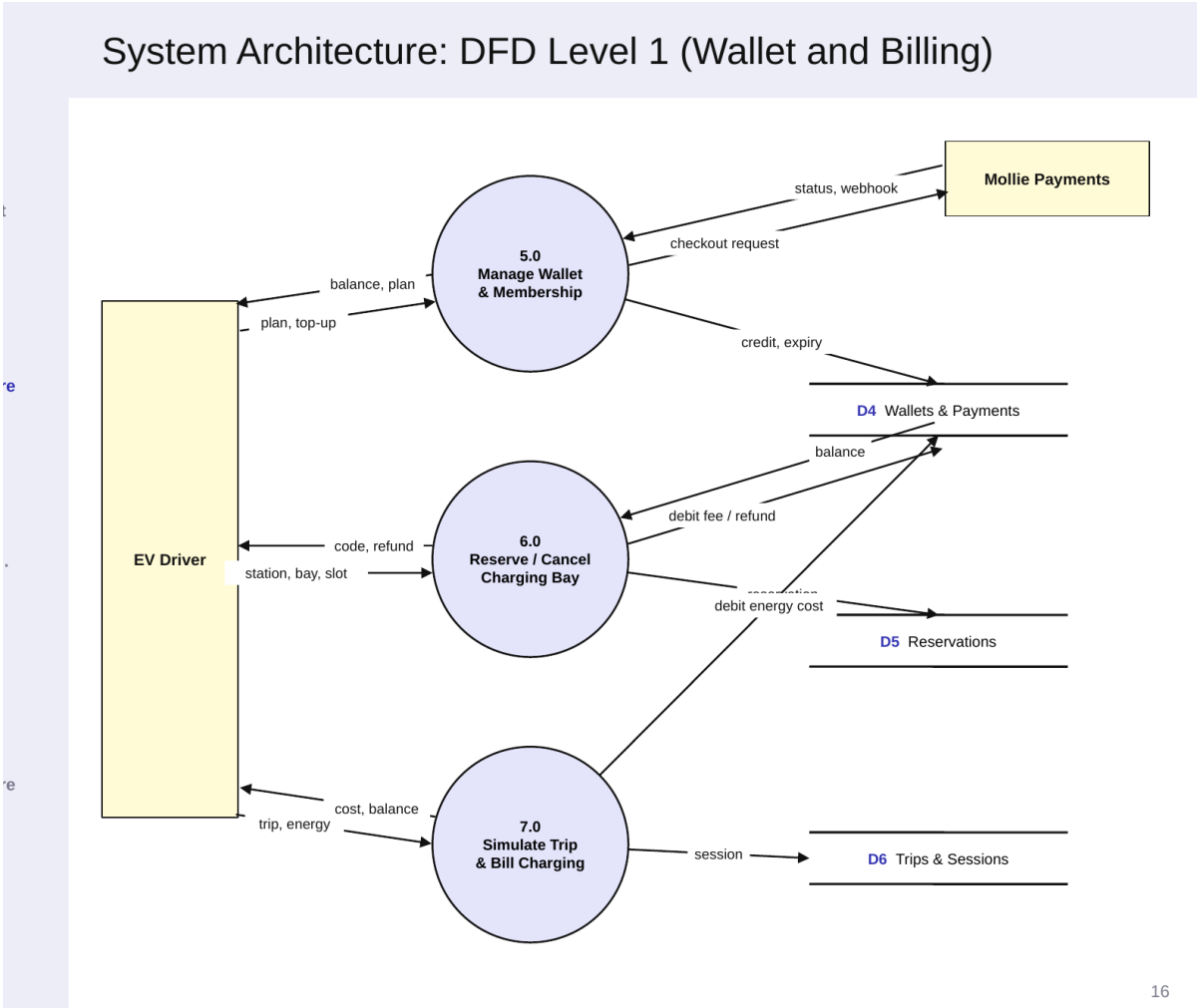


Figure 5.5: Level-1 DFD: Wallet and Billing

5.1.2.5 Level-1 Data Flow Diagram: Fleet Management

The fleet management data flow diagram describes the management of fleet vehicles, fleet reservations, vehicle telemetry, and charging-cost reports. Fleet owners can view registered fleet vehicles, monitor available telemetry, request charging reservations, and review fleet sessions and associated costs.

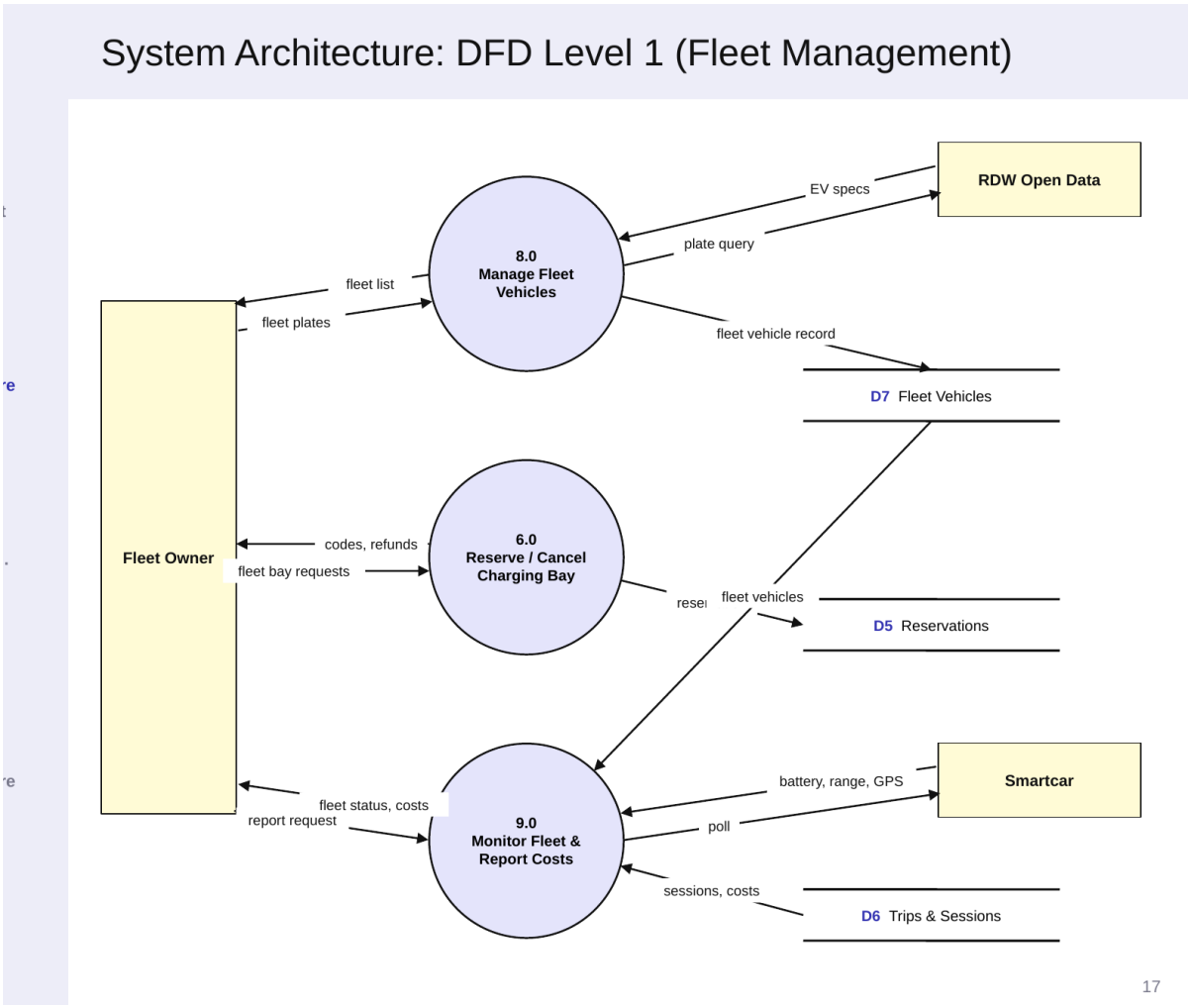


Figure 5.6: Level-1 DFD: Fleet Management

5.1.3 System Architecture

The TrustCharge architecture is organized into three main tiers: the presentation tier, application tier, and data tier.

5.1.3.1 Presentation Tier

The presentation tier is implemented using React and provides the user interface for authentication, vehicle verification, charging station discovery, AI-assisted recommendations, trip planning, reservations, wallet management, and memberships. Tailwind CSS supports interface styling, while Leaflet supports map-based station visualization.

5.1.3.2 Application Tier

The application tier uses Node.js and Express to implement the backend API and business logic. Its components include authentication, vehicle verification, station aggregation, reliability scoring, AI-assisted recommendations, route planning, wallet management, membership

processing, reservation management, and trip simulation. The backend communicates with external services through their respective APIs.

5.1.3.3 Data Tier

The data tier uses SQLite for persistent application data. It stores structured information related to users, vehicles, telemetry, station observations, ratings, reports, wallets, reservations, and trip sessions. Database constraints and transaction-handling mechanisms support data integrity.

5.1.3.4 External Services

TrustCharge integrates external services for authentication, vehicle verification, charging station information, vehicle telemetry, payments, AI-assisted recommendations, and navigation. These services include Firebase Authentication, RDW Open Data, Smartcar, NDW, TomTom, OpenChargeMap, Mollie, Google Gemini, and OpenStreetMap services.

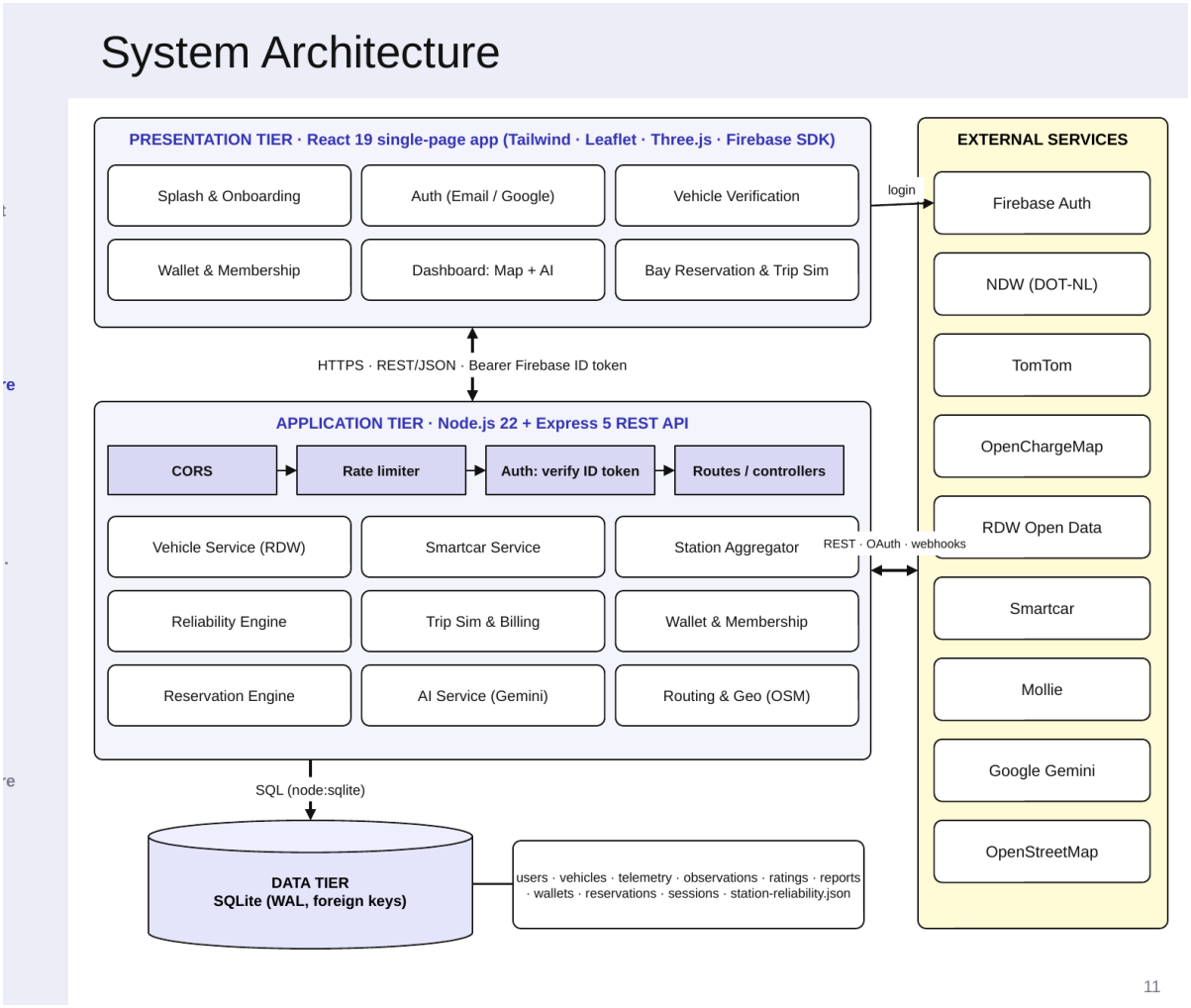


Figure 5.7: System Architecture of TrustCharge

5.2 Implementation

5.2.1 Web Application Development

The frontend is implemented using React, while the backend provides REST APIs through Node.js and Express. The application integrates charging station information, vehicle-related data, reservations, memberships, wallet operations, and AI-assisted recommendations through the corresponding backend services.

5.2.2 Charging Station Aggregation and Reliability

The station aggregation component collects available station information from the integrated external providers. The reliability component uses recorded station observations, operational status, ratings, and reports to calculate an observed reliability score. When sufficient evidence is unavailable, the system indicates that there is insufficient observed history rather than presenting an unsupported reliability score.

5.2.3 Reservation and Membership Management

The reservation component manages charging bay bookings according to configured membership limits and reservation rules. The wallet component maintains transaction records and supports applicable payments and refunds. Membership plans provide different reservation limits and benefits according to their configured rules.

5.2.4 AI-Assisted Recommendations and Navigation

The AI assistant uses Google Gemini to provide station recommendations from the station information supplied to it. OpenStreetMap services support route calculation, geocoding, and nearby-point-of-interest discovery. AI-based charger reliability prediction and queue-time prediction are considered future enhancements rather than completed capabilities.

5.2.5 Testing and Evaluation

The system is evaluated by testing authentication, vehicle verification, station data retrieval, reliability scoring, reservation management, wallet transactions, membership benefits, AI recommendations, and trip simulation. Integration testing is used to examine communication between the frontend, backend, database, and external services.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Experimental Set-up

TrustCharge was developed as a centralized electric vehicle (EV) charging platform that integrates charging station information, vehicle data, navigation, reservations, membership features, and payment workflows. The system uses a React-based frontend and a Node.js backend to connect the user interface with external services and the application database.

The experimental setup focuses on demonstrating the implemented features and evaluating the available station data. The platform aggregates information from multiple external services. However, real-time charging availability and AI-based reliability prediction have not been conclusively validated.

Component	**Technology Used**
Frontend	React.js and Tailwind CSS
Backend	Node.js and Express.js
Database	SQLite
Authentication	Firebase Authentication
AI Assistant	Google Gemini
Charging Station Data	RDW Open Data and OpenChargeMap
Vehicle Integration	Smartcar API
Navigation and Mapping	OpenStreetMap and TomTom APIs
Additional Data Sources	NDW Open Data
Payment Integration	Mollie payment gateway (test mode)

Table 6.1: Technology Stack Used in TrustCharge

6.2 Implemented Features

The major features implemented in TrustCharge are summarized in Table 6.2. These features demonstrate the integration of charging station discovery, user authentication, vehicle integration, navigation, and supporting workflows.

Sl. No.	Feature	Description
1	User Authentication	Allows users to access the platform through Firebase Authentication.
2	Charging Station Discovery	Aggregates charging station information from external data sources.
3	Vehicle Integration	Supports vehicle data integration through Smartcar, subject to authorization and API availability.
4	Navigation	Provides map-based station discovery and navigation features.
5	Reservation Management	Provides an interface for selecting and managing charging reservations.
6	Membership Management	Supports membership-related features and priority-access workflows.
7	Payment Workflow	Integrates Mollie in test mode to demonstrate the payment workflow.
8	AI Assistant	Integrates an AI assistant to support user queries.
9	Station Reliability	Presents station information and available observations to support reliability assessment.

Table 6.2: Summary of Implemented Features in TrustCharge

6.3 Station Data Aggregation

TrustCharge collects charging station information from multiple external services. The recorded project observations indicate that the platform tracks 1,574 charging stations and contains 3,990 observation samples collected through nine external services.

During the reported observation period, the observed station status values were recorded as

UNKNOWN. Therefore, the aggregated data demonstrates the collection and storage of station information, but it does not establish that real-time charger availability was successfully verified.

Parameter	**Observed Value**
Charging stations tracked	1,574
Observation samples	3,990
External services involved	9
Station status observations	UNKNOWN during the reported observation period
Real-time charging availability	Not established by the available observations
Historical reliability prediction	Requires sufficient validated historical data

Table 6.3: Summary of Charging Station Data Aggregation

6.4 Vehicle, Reservation, and Payment Workflows

The platform includes workflows for vehicle integration, reservation management, membership features, and payment processing. Vehicle integration depends on authorization, API permissions, and the availability of external vehicle data.

The reservation interface supports the reservation workflow, but successful execution of an external charging session has not been established. Similarly, the payment workflow was demonstrated using Mollie test mode. Production payment settlement has not been established.

Workflow	Implementation	Current Status
Vehicle Integration	Smartcar API integration	Depends on authorization, permissions, and API availability.
Reservation	Reservation interface and ticket workflow	Interface implemented; external charging-session execution is not established.
Membership	Membership-related functionality	Supports membership workflow demonstration.
Payment	Mollie payment gateway	Test-mode checkout demonstrated; production settlement is not established.
Navigation	Map-based station discovery and routing	Uses available mapping and navigation services.

Table 6.4: Summary of Vehicle, Reservation, and Payment Workflows

6.5 Trip Simulation and Navigation

The navigation component is designed to assist users in locating charging stations and exploring routes using available mapping services. It combines station information with map-based discovery to support trip planning.

The effectiveness of navigation depends on the availability and accuracy of external map and station data. Since real-time charging availability was not established during the reported observation period, the platform should not be considered to have verified that a selected station will have an available charger when a user arrives.

6.6 Evaluation and Limitations

The current evaluation focuses on the demonstrated application workflows and the available station data. The results indicate that station aggregation and several supporting interfaces have been implemented. However, the reliability of live station status, production payment settlement, and external charging-session execution has not been conclusively established.

AI-based reliability prediction and queue-time estimation remain future development objec-

tives. These functions require sufficient validated historical observations and systematic evaluation before their prediction performance can be reported.

Sl. No.	Evaluation Item	Result or Limitation
1	Station Data Aggregation	Tracking of 1,574 stations and 3,990 observation samples was recorded.
2	Real-Time Availability	Actual charger availability was not established by the recorded observations.
3	Reliability Assessment	Depends on the availability and quality of station observations and reports.
4	AI-Based Prediction	Reliability and queue-time prediction remain future development objectives.
5	Reservation Workflow	Reservation interface is available; external charging-session execution has not been established.
6	Payment Workflow	Mollie test-mode checkout was demonstrated; production settlement has not been established.
7	Vehicle Integration	Functionality depends on external API permissions and data availability.
8	Navigation	Uses available mapping and navigation services.

Table 6.5: Evaluation Status and Limitations of TrustCharge

6.7 Discussion

The implementation of TrustCharge demonstrates an approach to centralizing electric vehicle charging station information and providing supporting services through a unified platform. The application combines a frontend interface, backend services, a database, and external data integrations to support station discovery and related user workflows.

The recorded station aggregation results indicate that the platform has collected information about 1,574 charging stations and 3,990 observation samples from nine external services.

However, the UNKNOWN status values recorded during the reported period limit the conclusions that can be drawn about actual charger availability. More reliable status observations are needed to evaluate the effectiveness of real-time station monitoring.

The vehicle, reservation, and payment workflows demonstrate the intended structure of the application. Nevertheless, vehicle functionality depends on external authorization and data availability, the reservation interface does not by itself establish successful execution of a charging session, and test-mode payment processing does not establish production settlement.

The proposed AI-based reliability and queue-time prediction features could help users make better charging decisions. However, their effectiveness cannot be claimed without sufficient historical data, a clearly defined prediction method, and quantitative evaluation against observed outcomes.

Future development should focus on improving station-status verification, collecting reliable historical observations, evaluating external API integrations, testing the complete reservation workflow, and validating payment processing beyond test mode. Once sufficient data becomes available, reliability and queue-time prediction models can be developed and evaluated using appropriate performance metrics.

Overall, TrustCharge provides a foundation for a centralized EV charging platform. Further implementation and systematic testing are required to establish its reliability, operational effectiveness, and suitability for real-world use.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

TrustCharge is a centralized web platform intended to simplify the process of finding and using electric-vehicle charging infrastructure. It combines station discovery from multiple sources with vehicle verification, optional telemetry, map-based navigation, bay reservation, wallet and membership workflows, trip simulation, and an AI assistant. The project uses a React and Tailwind CSS frontend with a Node.js and Express backend and SQLite persistence, alongside external services for identity, payment, mapping, station data, vehicle information, and AI-assisted interaction.

The project evidence demonstrates a working set of integrated user-facing workflows. At the same time, the available observations do not establish validated live status for every station, and the trained AI models for charger reliability and queue-time prediction remain future work. The report therefore distinguishes implemented functionality from proposed predictive capabilities. This distinction is important for presenting the current system accurately and for defining a measurable next stage of development.

7.2 Future Scope

Future work can extend TrustCharge in the following directions:

- Develop a labelled historical dataset from station status, connector availability, faults, user reports, and timestamps.
- Train and validate charger-reliability models using precision, recall, F1-score, and calibration measures where suitable ground truth is available.
- Develop waiting-time and occupancy prediction models and evaluate them using MAE, RMSE, and MAPE against appropriate baselines.
- Incorporate traffic conditions, station utilization, charging power, time of day, and local demand into station ranking and trip recommendations.

- Improve data freshness monitoring, provider-failure handling, source attribution, and user-visible status confidence.
- Conduct field testing across different charging networks and regions to assess scalability, usability, and the reliability of recommendations.
- Strengthen security review, privacy controls, payment-flow verification, accessibility, and mobile responsiveness before production deployment.

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